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Multiple Flooding locations in Oneida County, NY in 2017: An approach to determine flood vulnerable sites using LiDAR in Geographical Information Systems (GIS) and flood simulations

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Introduction

Oneida County is in central Upstate New York. In late June and early July of 2017, areas of Oneida County within the Mohawk Valley experienced heavy rainfall from numerous storms, which affected much of Upstate New York. This heavy rainfall caused excess overflow of the Mohawk River and the creeks branching from it, such as the 17-mile Sauquoit Creek in the southern part of the county, leading to major flooding of the surrounding area, which occurred on July 1st, 2017. Homes, buildings, and roads were damaged, and catch basins collapsed. Whitesboro, one of the towns in which the Sauquoit Creek and Mohawk River run through, as well as Utica, were just two of the towns in the county that experienced some of the worst of the flooding, with Whitesboro receiving about 3.38 inches of rain and Utica receiving 4.01 inches in the 24 hours prior (weather.gov). Oneida County proceeded to declare a state of emergency for a few days. Since this was not the first flood that Oneida County has experienced and most likely will not be the last, the State and County have begun to fund projects to widen and deepen the Sauquoit Creek, which would potentially fix the recurrent flooding problems. Previous research has explored the spatial analysis and flood simulations at the nearby regions (Swan et al., 2016), and the purpose of this study is to determine the areas most vulnerable to groundwater and river flooding around the Mohawk River and Sauquoit Creek using LiDAR and GIS water flood simulations and data as an efficient method to help identifying vulnerable locations of flood in the future.

Methodology

The first method that was employed to begin collecting data was finding pictures of the flood using Google. By obtaining multiple different views and locations of the flood, the actual level of water surface of the flood and destruction was more easily calculated. The water level of flood of each area was obtained by approximating the elevation that was given by Google Earth Pro and then adding the water level to this number. This flooding event happened to be widespread throughout several different towns in Oneida County and also set many flood records, and therefore it was very well documented by the residents as well as several different news agencies. The photos were obtained through various news websites and showed the levels of flood water in different areas in Oneida County. Then, each photograph location was identified using Google Earth Pro Street View in order to save pinpoints of the exact locations. Screenshots were also taken of each photograph location in Street View in order to more accurately compare the flood levels to the normal view of the area. In all, twenty-three different test locations were able to be documented in both flood photographs and Google earth photographs. Once selected, each site's coordinates were collected, and the flood heights recorded using Google Earth's elevation and the flood pictures (Table 1).

First, each of the saved site points from Google Earth was saved as an individual .kmz file and then imported into Global Mapper. The information tool on Global Mapper was used to obtain file names of the .las files that corresponded with the LiDAR data of the affected area. Both the .kmz and existing .las data were then imported into Global Mapper thereby creating a map of the area with the points overlaid.

Table 1: Comparison of the elevation data derived from Google Earth and the LiDAR-DEM, flood height from both LiDAR and Google Earth are in feet, and differences (DIF column in feet) in those two heights for each of the twenty-three test sites. Average is 1.6, standard deviation is 1.5.

Google Earth							
	Name	(ft)	Latitude	Longitude	Lidar-DSM (ft)	LiDAR Data (m)	DIF(ft)
1	JAY-K Lumber	502	43°04'54.98" N	75°19'28.17" W	505.0	153.9	3
2	Highway	444	43°06'00.32" N	75°14'40.10" W	445.5	135.8	1.5
3	Genese Street	530	43°04'55.91" N	75°16'12.57" W	530.5	161.7	0.5
4	Lincoln/Noyes	451	43°05'49.90" N	75°14'58.09" W	454.0	138.4	3
5	Highway Exit	448	43°06'03.12" N	75°14'32.63" W	443.0	135.0	5
6	EZ Bottle	446	43°05'55.64" N	75°13'10.01" W	446.0	135.9	0
7	Fire House Road	453	43°07'30.48" N	75°12'20.38" W	454.0	138.4	1.017
8	Hannaford	612	43°03'16.40" N	75°16'36.08" W	617.0	188.1	5
9	Dudley Ave	611	43°04'55.87" N	75°14'19.65" W	611.0	186.2	0
10	Panella's	459	43°05'54.49" N	75°18'11.46" W	459.0	139.9	0
11	Wilbur Road	532	43°04'41.79" N	75°17'55.77" W	533.6	162.7	1.643
12	Riverside Center	441	43°07'40.00" N	75°13'16.58" W	441.9	134.7	0.92
13	Medicine Shoppe	424	43°06'50.07" N	75°16'25.00" W	424.0	129.2	0
14	Brookline Drive	540	43°04'18.66" N	75°16'57.83" W	541.6	165.1	1.6
15	Burger King	419	43°06'58.84" N	75°17'23.75" W	421.5	128.5	2.5
16	Whitesboro Aerial	418	43°07'05.27" N	75°17'16.45" W	418.4	127.5	0.376
17	Whitesboro Aerial 2	416	43°07'08.95" N	75°17'13.25" W	416.3	126.9	0.25
18	Trailer Park Aerial	420	43°07'02.58" N	75°17'12.27" W	418.3	127.5	1.75
19	Whitesboro Aerial 3	411	43°07'11.74" N	75°17'06.91" W	413.2	125.9	2.213
20	Whitesboro Aerial 4	415	43°07'12.49" N	75°17'11.44" W	414.5	126.3	0.5
21	Whitesboro Aerial 5	415	43°07'09.31" N	75°17'15.57" W	416.5	126.9	1.5
22	Whitesboro Aerial 6	408	43°07'09.32" N	75°16'59.20" W	411.7	125.5	3.735
23	Whitesboro Aerial 7	415	43°07'02.63" N	75°17'02.09" W	413.2	125.9	1.787

Once the map was complete, the test site points overlaid the elevation map so that the elevation of each point was easily seen. Then the program was used to generate a 3D version of the map utilizing a DEM obtained from LiDAR points in order to view the test area at every possible angle, which makes it much easier to view the flood simulation that is included in the program. Once the simulation began, each test site was looked at individually and the water level was changed until the water reached the same level or at an approximate level at the site in the program as it had in the actual flood. Once the levels visually matched up, the measured level of the water in the program was then recorded in Table 1. Once all sites had been recorded, the same flood program was also used in the 2D version of Global Mapper, but this time it simulated a coverage zone of flooded areas of test sites 13 and 15-23. The normal elevation of the river was then inputted into the system, 124 meters for this local area, and then the difference from that to the other sites, about 4 m higher, was inputted at the flood height. The flood height was changed various times until the most accurate flood map was achieved. This was done in both the bare earth model (only ground points) and the all included LiDAR points model for comparison.

Results

Using the LiDAR data and flood simulation, the flood map of the Whitesboro area was determined by using the elevation of the river at 124 m and the flood level at 4.5 m. The area that seemed to be the most affected by the flood was the large cluster of points near Whitesboro, which were points 15-23. This area is close to the river and even the points that were farther were affected by the resulting rush of groundwater from various creeks that split off of the river. The LiDAR data were much more effective in comparing the amount of flooding since the simulation immediately gave exact numbers there is more confidence in those results. Comparatively the data collected from Google Earth had to be approximated which allows for human error to inhibit the results. Furthermore, using the bare earth model in global mapper did not yield effective results since it did not take into account the existing various objects on the ground surface that affected the flow of the groundwater. Discerning which method was best, was easier once all the data was entered into a table; the comparison was clearer.

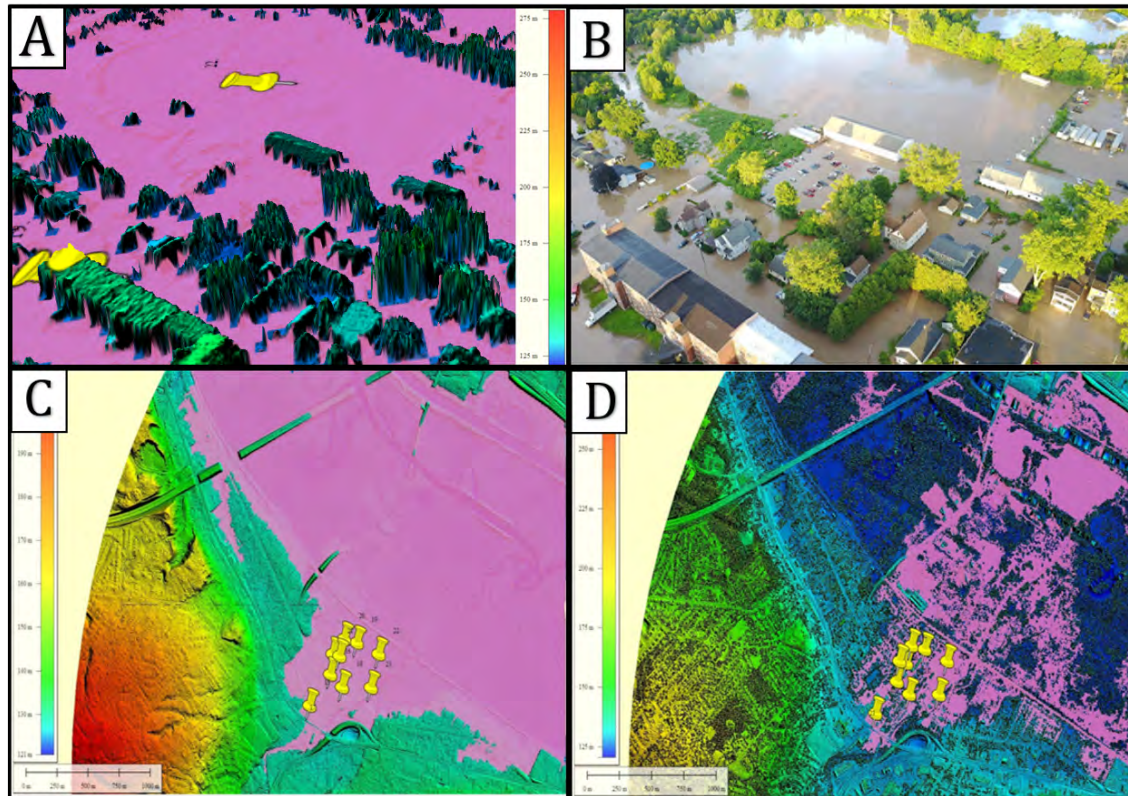


Figure 1: (A) Test Site 22, 3D LiDAR DSM flood simulation. (B) Test Site 22, aerial photograph of actual flooding, July 1, 2017 (David Olney and Thomas Sheehy). (C) LiDAR bare earth flood coverage model of test sites 13, 15-23. (D) All LiDAR data flood coverage model of test sites 13, 15-23.

Discussions

It was determined that only points 13 and 15-23 would be best to analyze as a local region affected by the same flood cause for this research. While the other sites did experience significant flooding from this event, it was found to not have been caused by groundwater or river flooding, but rather from either poor drainage, runoff from a higher elevation, or multiple other creeks. It was decided that sites 13, 15-23 in the Whitesboro area were all affected by groundwater and river flooding, and all from the Mohawk River and Sauquoit Creek, which made for a selective study area with more accurate results. These sites also seemed to have received the more intense amount of flooding. This seemed to be caused by the river flooding and then the access water caused the creeks branching off of the river to overflow. The vast majority of groundwater flooding seemed to be due to the various creeks overflowing as a result of the initial overflow of the river. A few other points had a large amount of flooding, but these were not due to sheer amount of groundwater present. In a couple of cases there was simply poor drainage in the area this was the case with points two, five and six. In other cases, the area was simply low-lying and therefore water was able to collect. While the Google Earth data was low resolution it did give a starting point for further analysis with the LiDAR data using simulations in Global Mapper. LiDAR data could have further use in both predicting floods and identifying high risk areas. It is also worth noting that obtaining images from Google Earth and other media for the purpose of extracting flood data saves time and expenses that would be incurred if observations in the field were necessary. This method enables quick response and effective decision actions when an area becomes at risk, and gleaming the highest possible water level will enable rescue efforts to be both faster and more effective. If a 3D simulation is done with the LiDAR data, it is simply to study different clusters of points based on the supposed cause of the flood, which differs typically depending on location in respect to the river. The height of the flood could change dramatically based on various factors of the area; responders could make better decisions based on the increased amount of information. Also, there is less time needed to evaluate images since once a simulation of the flood in question has been run, the necessary information is easy to obtain. There are numerous applications of geohazard and

environmental impact during flooding using this methodology. For instance, contamination transpires in undesirable areas due to flooding, therefore predicting and mapping future floods can assist in potential issues regarding contamination.

It was also determined that based upon the comparison of the bare earth model and the all included model of the flood coverage map overlaid with the test points, the bare earth model gave a less accurate flood map. This is because with the bare earth model, there are no objects in the data to block or constrain the path of the water and divert it to other areas, but with the all the LiDAR points included model, all above ground objects included in the LiDAR data are also included in the simulation, and therefore it takes into account the objects on the ground when calculating where the water will go (water flow simulation) as it rises and giving a much more accurate flood coverage map. This can also be seen in the side by side picture comparison in figure 2, where it can be seen that the model with all the data (D) is much more detailed than the bare earth model (C), and that the test points are more accurately covered in (D).

Conclusion

This method of combining LiDAR and GIS simulations and photographic flood data can be an accurate way to predict and map current floods in order to mitigate the losses of locals in similar floods, and to better identify and prepare at risk areas especially in emergency situations.

References:

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Poster Presentation